# Weekly Progress Report

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Domain: JAVA

Date of submission: 24-03-2026

**Week Ending: 01**

**1.Banking Information System**

**I. Overview:**

This week, the primary focus was on understanding the Core Java internship structure and developing a Banking Information System project. Efforts were made to learn and apply fundamental Java concepts while implementing the core functionalities of the banking system. Additionally, learning resources were utilized to improve programming skills and problem-solving abilities.

**II. Achievements:**

**1. Core Java Familiarization:**

- Explored Core Java concepts such as classes, objects, methods, and control structures.
- Gained a clear understanding of object-oriented programming principles.
- Successfully implemented basic Java programs to strengthen logic-building skills.

**2. Project Development (Banking Information System):**

**Name of the project:** Banking Information System

- Developed core functionalities such as user registration and login system.
- Implemented deposit, withdrawal, and fund transfer features.
- Added transaction history (account statement) functionality.
- Designed a menu-driven console interface for user interaction.
- Successfully executed and tested the project with multiple user scenarios.

**3. Skill Enhancement:**

- Improved knowledge in Java collections (HashMap, ArrayList).
- Learned error handling techniques such as checking insufficient balance and invalid inputs.
- Enhanced debugging and logical thinking skills through hands-on coding.

**III. Challenges:**

**1. Project Implementation:**

- Faced difficulty in structuring the project and dividing it into classes (Account, Bank, Main).
- Encountered errors during compilation and method implementation.

**2. Logic Building:**

- Initially found it challenging to implement fund transfer and manage multiple user accounts.
- Required additional practice to understand how data flows between classes.

**IV. Learning Resources:**

**1. Java Documentation & Materials:**

- Referred to Core Java notes and online documentation to understand concepts.
- Used examples and tutorials to implement real-time project features.

**2. Online Learning Platforms:**

- Practiced Java programs from coding platforms and tutorials.
- Watched video tutorials to understand project implementation step-by-step.

**V. Next Week's Goals:**

**1. Project Enhancement:**

- Implement file handling to store user data permanently.
- Add date and time to transaction history.
- Improve the user interface for better usability.

**2. Skill Development:**

- Learn advanced Java concepts and improve coding efficiency.
- Work on optimizing the project and adding additional features.

**VI. Additional Comments:**

This week provided a strong foundation in Core Java and practical project development. The Banking Information System helped in understanding real-world application development and improved confidence in coding and debugging. Further enhancements will make the project more robust and user-friendly.

**Code:**

**GITHUB:**

**https://github.com/Jaishree935/JAVA_LongLevelCoding/tree/main/BANKING%20APPLICATION**

**Output:**

```
PS C:\Users\jaish\Desktop\INTERN_JAVA\Banking Application> javac *.java
PS C:\Users\jaish\Desktop\INTERN_JAVA\Banking Application> java Main

 1.Register 2.Login 3.Exit
 1
 Enter Name: Jaishree
 Set Password: Jaiz123
 Initial Deposit: 10000
 Account Created. Acc No: 1001

 1.Register 2.Login 3.Exit
 2
 Account No: 1001
 Password: Jaiz123
 Login Success

 1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
 1
 Amount: 5000

 1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
 4
 Transactions:
 Deposited: 5000.0
```

```
1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
5
 Current Balance: 15000.0

1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
6

1.Register 2.Login 3.Exit
1
Enter Name: Abi
Set Password: Abi123
Initial Deposit: 50000
Account Created. Acc No: 1002

1.Register 2.Login 3.Exit
2
Account No: 1002
Password: Abi123
Login Success

1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
3
Receiver Acc No: 1001
Amount: 200
```

```
1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
3
Receiver Acc No: 1001
Amount: 20
Transfer Successful

1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
4
Transactions:
Transferred: 200.0 to 1001
Transferred: 20.0 to 1001

1.Deposit 2.Withdraw 3.Transfer 4.Statement 5.Balance 6.Logout
6

1.Register 2.Login 3.Exit
3
```

2. **Music player application**

**I. Overview:**

This week, the primary focus was on understanding the Core Java internship structure and developing a Music Player Application. Efforts were made to learn and apply Java programming concepts while implementing essential functionalities such as music playback, playlist management, and library organization. Additionally, various learning resources were used to enhance programming and problem-solving skills.

**II. Achievements:**

**1. Core Java Familiarization:**

• Explored Core Java concepts such as classes, objects, methods, and control structures.

• Strengthened understanding of object-oriented programming principles.

• Practiced basic Java programs to improve logical thinking and coding skills.

**2. Project Development (Music Player Application):**

Name of the project: Music Player Application

• Developed core functionalities such as play, pause, resume, and stop music playback.

• Implemented a music library to store and display available songs.

• Created playlist management features to add, view, and organize songs.
• Added search functionality to find songs by name.
• Implemented volume control for audio adjustment.
• Designed a menu-driven console interface for smooth user interaction.
• Successfully executed and tested the application with multiple scenarios.

**3. Skill Enhancement:**

•Improved knowledge in Java collections such as ArrayList.
• Learned how to manage dynamic data like song lists and playlists.
• Enhanced debugging and logical thinking skills through project development.
• Gained experience in designing real-time applications using Java.

**III. Challenges:**

**1. Project Implementation:**

• Faced difficulty in organizing the project structure and managing multiple functionalities.
• Encountered issues while implementing playlist management and navigation features.

**2. Logic Building:**

• Initially found it challenging to manage song selection and indexing correctly.
• Required additional effort to implement search and playlist features efficiently.

**IV. Learning Resources:**

**1. Java Documentation & Materials:**

• Referred to Core Java documentation and notes to understand concepts.
• Used example programs to implement real-time functionalities.

**2. Online Learning Platforms:**

• Practiced coding through online tutorials and programming platforms.
• Watched video tutorials to understand the working of music player applications.

**V. Next Week's Goals:**

**1. Project Enhancement:**

• Implement file handling to import and store real music files.
• Improve the user interface using Java Swing or JavaFX.
• Add advanced features such as shuffle, repeat, and better navigation.

**2. Skill Development:**

- Learn advanced Java concepts and improve coding efficiency.
- Work on enhancing the project with additional real-world features.

**VI. Additional Comments:**

This week helped in gaining practical experience in developing a Music Player Application using Core Java. The project improved understanding of real-world application design, data handling, and user interaction. It also enhanced problem-solving and debugging skills. Future improvements will make the application more interactive and feature-rich.

**Code:**

**GITHUB:**

**https://github.com/Jaishree935/JAVA_LongLevelCoding/tree/main/MUSIC%20PLAYER%20APPLICATION**

**Output:**

```
PS C:\Users\jaish\Desktop\INTERN_JAVA\Music player application> java Main

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
1

 Music Library:
1 . Song1 - AR Rahman (Album1)
2 . Song2 - Anirudh (Album2)
3 . Song3 - Yuvan (Album3)

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
2
Enter Song Number: 1
Playing: Song1 - AR Rahman (Album1)

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
3
Paused
```

```
 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
4
Resumed: Song1 - AR Rahman (Album1)

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
5
Stopped

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
6
Enter Song Number: 2
Added to playlist

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
7

 Playlist
Song2 - Anirudh (Album2)
```

```
 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
8
Playlist Shuffled

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
9
Enter Keyword: Song1

 Search Results:
Song1 - AR Rahman (Album1)

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
10
Enter volume (0-100): 80
Volume set to: 80

 1.Show Library 2.Play 3.Pause 4.Resume 5.Stop
6.Add Playlist 7.Show Playlist 8.Shuffle
9.Search 10.Volume 11.Exit
11
```

## 3. Human Resource Management System

**I. Overview:**

This week, the primary focus was on understanding the Core Java internship structure and developing a Human Resource Management System (HRMS). Efforts were made to learn and apply Java programming concepts while implementing functionalities such as employee management, attendance tracking, and leave management. Additionally, learning resources were utilized to improve coding skills and problem-solving abilities.

**II. Achievements:**

**1. Core Java Familiarization:**

• Explored Core Java concepts such as classes, objects, methods, and control structures.

• Gained a clear understanding of object-oriented programming principles.

• Practiced basic Java programs to strengthen logical thinking and coding skills.

**2. Project Development (Human Resource Management System):**

**Name of the project: Human Resource Management System (HRMS)**

• Developed employee management functionalities such as adding, viewing, updating, and deleting employee records.

• Implemented attendance tracking system to mark employees as present or absent.

• Created leave management functionality to apply and view leave records.

• Added search functionality to find employees by ID.

• Designed a secure login system for admin authentication.
• Developed a menu-driven console interface for easy user interaction.
• Successfully executed and tested the system with multiple employee records.

**3. Skill Enhancement:**

• Improved knowledge in Java collections such as HashMap for storing employee data.
• Learned how to manage and organize multiple records dynamically.
• Enhanced debugging and problem-solving skills through project development.
• Gained practical experience in developing real-world management systems.

**III. Challenges:**

**1. Project Implementation:**

• Faced difficulty in structuring the project and managing multiple modules like employee, attendance, and leave.
• Encountered errors while implementing CRUD operations and data storage.

**2. Logic Building:**

• Initially found it challenging to manage multiple employee records efficiently.
• Required additional practice to implement search and attendance tracking features correctly.

**IV. Learning Resources:**

**1. Java Documentation & Materials:**

• Referred to Core Java notes and documentation to understand key concepts.
• Used example programs to implement real-time functionalities.

**2. Online Learning Platforms:**

• Practiced coding through tutorials and programming platforms.
• Watched video tutorials to understand HRMS system implementation.

**V. Next Week's Goals:**

**1. Project Enhancement:**

• Implement file handling to store employee data permanently.
• Add date-based attendance tracking.
• Improve user interface using Java Swing or JavaFX.

**2. Skill Development:**

• Learn advanced Java concepts and improve coding efficiency.
• Enhance the project with additional features like reports generation.

**VI. Additional Comments:**

This week provided valuable experience in developing a Human Resource Management System using Core Java. The project helped in understanding real-world data management, user authentication, and system design. It also improved confidence in coding and debugging. Future enhancements will make the system more efficient and user-friendly.

**CODE:**

**GITHUB:**

**https://github.com/Jaishree935/JAVA_LongLevelCoding/tree/main/Human%20Resource%20Management%20System%20**

**OUTPUT:**

```
javac *.java
PS C:\Users\jaish\Desktop\INTERN_J
java Main
Username: admin
Password: 123

1.Add 2.View 3.Update 4.Delete
5.Attendance 6.View Attendance
7.Leave 8.View Leave
9.Search 10.Exit
1
ID: 101
Name: Jai
Dept: IT
Role: Developer
Contact: 9875424561
? Employee Added

1.Add 2.View 3.Update 4.Delete
5.Attendance 6.View Attendance
7.Leave 8.View Leave
9.Search 10.Exit
2
```

```
101 | Jai | IT | Developer | 9875424561

1.Add 2.View 3.Update 4.Delete
5.Attendance 6.View Attendance
7.Leave 8.View Leave
9.Search 10.Exit
3
Enter ID: 101
Kumar
? Updated

1.Add 2.View 3.Update 4.Delete
5.Attendance 6.View Attendance
7.Leave 8.View Leave
9.Search 10.Exit
5
ID: 101
Status (Present/Absent): Present
? Attendance marked
```

```
1.Add 2.View 3.Update 4.Delete
5.Attendance 6.View Attendance
7.Leave 8.View Leave
9.Search 10.Exit
6
EmpID: 101 -> Present
```

## 4. Console-based expense tracker application

**I. Overview:**

This week, the primary focus was on understanding the Core Java internship structure and developing a Console-based Expense Tracker Application. Efforts were made to learn and apply Java programming concepts while implementing functionalities such as expense recording, category management, and expense tracking. Additionally, various learning resources were utilized to enhance coding and problem-solving skills.

**II. Achievements:**

**1. Core Java Familiarization:**

• Explored Core Java concepts such as classes, objects, methods, and control structures.
• Gained a clear understanding of object-oriented programming principles.
• Practiced basic Java programs to improve logical thinking and coding skills.

**2. Project Development (Expense Tracker Application):**

**Name of the project: Console-based Expense Tracker Application**

• Developed functionality to record expenses with details such as date, amount, category, and description.
• Implemented category management to organize expenses effectively.
• Created features to view and track all recorded expenses.
• Added filtering functionality to view expenses based on category.
• Implemented update and delete operations for managing expenses.
• Developed a feature to calculate total expenses for better financial tracking.
• Designed a menu-driven console interface for smooth user interaction.
• Successfully executed and tested the application with multiple expense records.

**3. Skill Enhancement:**

• Improved knowledge in Java collections such as ArrayList and HashSet.
• Learned how to handle dynamic data like expense records and categories.
• Enhanced debugging and problem-solving skills through real-time coding.
• Gained practical experience in developing a real-world utility application.

**III. Challenges:**

**1. Project Implementation:**

• Faced difficulty in managing multiple expense records efficiently.
• Encountered errors while implementing update and delete functionalities.

**2. Logic Building:**

• Initially found it challenging to implement filtering and category-based tracking.
• Required additional effort to maintain proper data structure for expenses.

**IV. Learning Resources:**

**1. Java Documentation & Materials:**

• Referred to Core Java notes and documentation to understand key concepts.
• Used example programs to implement application features.

**2. Online Learning Platforms:**

• Practiced coding through tutorials and programming platforms.
• Watched video tutorials to understand expense tracking system implementation.

**V. Next Week's Goals:**

**1. Project Enhancement:**

• Implement file handling to store expense data permanently.
• Add date-based filtering for better analysis.
• Improve user interface for better usability.

**2. Skill Development:**

• Learn advanced Java concepts and improve coding efficiency.
• Enhance the project by adding report generation features.

**VI. Additional Comments:**

This week provided valuable experience in developing a Console-based Expense Tracker Application using Core Java. The project helped in understanding real-world data management, user interaction, and application design. It also improved confidence in coding and debugging. Future enhancements will make the application more efficient and user-friendly.

**CODE:**

**GITHUB:**
**https://github.com/Jaishree935/JAVA_LongLevelCoding/tree/main/Console-based%20expense%20tracker%20application%20**

## OUTPUT:

```
PS C:\Users\jaish\Desktop\INTERN_JAVA\CONSOLE BASED>javac *.java
PS C:\Users\jaish\Desktop\INTERN_JAVA\CONSOLE BASED>java Main

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
1
Date: 24-03-026
Amount: 500
Category: Food
Description: Lunch
? Expense Added

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
1
Date: 24-03-2026
Amount: 200
Category: Travel
Description: Bus
? Expense Added
```

```
1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
2
1 | 24-03-026 | 500.0 | Food | Lunch
2 | 24-03-2026 | 200.0 | Travel | Bus

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
5
Enter Category: Food
1 | 24-03-026 | 500.0 | Food | Lunch

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
6
? Total Expense: 700.0

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
4
Enter ID: 1
New Amount: 600
? Updated
```

```
1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
3
Enter ID: 2
? Deleted

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
7
? Categories:
Travel
Food

1.Add 2.View 3.Delete 4.Update
5.Filter 6.Total 7.Categories 8.Exit
8
```